what can be the general proof behind this, so it is proven for all cases? does it exist? or no, is it there exists only

\item[27. ] Suppose the third column of $B$ is the sum of the first two columns. What can you say about the third column of $AB$?
\begin{itemize}
\item {\color{purple} If $B$ has dependence, then it is not invertible, thus it follows that $AB$ will also be dependent and singular because $det(B) = 0 \implies det(AB) =0$
\item We can prove that $AB$ third column will be $A(b_1) + A(b_2)$ because $B_3$ third column is $b_3 = b_1 + b_2$.
\item We can also state that there will be a lower dimension output.

**TL;DR (robot mode):**

Yes, there is a *general proof* that works for **all cases**. It does **not** depend on determinants or invertibility. It follows directly from **linearity of matrix multiplication**.
The third column of $AB$ is the sum of the first two columns of $AB$.

---

## Clean general proof (all cases)

Let

$$B = [\, b_1 \ b_2 \ b_3 \,], \quad \text{with } b_3 = b_1 + b_2$$

Matrix multiplication by $A$ acts **column-wise**:

$$AB = [\, Ab_1 \ Ab_2 \ Ab_3 \,]$$

Substitute $b_3$:

$$Ab_3 = A(b_1 + b_2)$$

By linearity of matrix multiplication:

$$A(b_1 + b_2) = Ab_1 + Ab_2$$

Therefore:

$$\boxed{\text{column}_3(AB) = \text{column}_1(AB) + \text{column}_2(AB)}$$

This holds **for every matrix $A$**, with no assumptions on invertibility.

---

### What you can say definitively

1. **Column dependence is preserved**
   If columns of $B$ are linearly dependent, then columns of $AB$ are also dependent.
   (Reason: linear maps preserve linear relations.)
2. **Determinants are unnecessary**
   Your determinant argument is *true but overkill* and fails if $B$ is not square.
   Column dependence alone is sufficient.
3. **"Lower dimension output" is imprecise**
   Correct statement: the **column space of $AB$** has dimension $\leq$ column space of $B$.
   Not necessarily strictly lower.

---

### Corrected concise answer (exam-ready)

> If the third column of $B$ equals the sum of the first two, then the third column of $AB$ equals the sum of the first two columns of $AB$. This follows from linearity of matrix multiplication. Hence, $AB$ has linearly dependent columns.

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### Sources



multiplication
- Axler, *Linear Algebra Done Right*, Ch. 2 — Linear maps preserve linear dependence
- Lay, Lay, McDonald, *Linear Algebra and Its Applications*, Section 2.1 — Matrix-vector products and column interpretation

If you want, I can rewrite your purple solution in fully correct, minimal form.

Note that you only proved it for the dimension of R^3, I am saying for R^m

**TL;DR (robot mode):**
The proof already works for $\mathbb{R}^m$ with no changes. It is **dimension-free**. Nothing in the argument uses $m = 3$.

---

## General statement (any $m$)

Let

$$B \in \mathbb{R}^{m \times n}, \quad A \in \mathbb{R}^{k \times m}$$

Write the columns of $B$ as

$$B = [\, b_1 \; b_2 \; \dots \; b_n \,], \quad b_i \in \mathbb{R}^m$$

Assume

$$b_3 = b_1 + b_2$$

---

## Column-wise multiplication (definition)

Matrix multiplication satisfies, for **any** column $j$:

$$(AB)_j = Ab_j$$

This is true for all $m$ (definition of matrix multiplication).

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### Apply linearity (dimension-free)

$$(AB)_3 = Ab_3 = A(b_1 + b_2)$$

By linearity of linear maps $A : \mathbb{R}^m \to \mathbb{R}^k$:

$$A(b_1 + b_2) = Ab_1 + Ab_2$$

Thus:

$$\boxed{(AB)_3 = (AB)_1 + (AB)_2}$$

---

### What this proves (all $m$)

- Column dependence in $B$ ⇒ column dependence in $AB$
- Holds for **any** $m$, any $k$, any $n$
- No determinants
- No invertibility assumptions
- No special role of $\mathbb{R}^3$

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### Key principle (the real theorem)

> Matrix multiplication by $A$ is a linear transformation $A : \mathbb{R}^m \to \mathbb{R}^k$, and linear transformations preserve linear relations.

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### Sources

- Axler, *Linear Algebra Done Right*, Ch. 2 — Linear maps preserve linear dependence
- Lay, Lay, McDonald, *Linear Algebra and Its Applications*, §2.1 — Column interpretation of $AB$
- Strang, *Linear Algebra and Its Applications*, §2.3 — Linearity of matrix multiplication